



Vaxcel® CEFTAZIDIME INJECTION

COMPOSITION:

Each vial contains Ceftazidime Pentahydrate (buffered with Sodium Carbonate) equivalent to Ceftazidime 1g, and 2g.

PRESENTATION:

White to cream-coloured, crystalline powder.

PHARMACOLOGY:

The absorption and elimination of ceftazidime were directly proportional to the size of the dose. The half-life following IV administration was approximately 1.9 hours. Less than 10% of ceftazidime was protein bound. The degree of protein binding was independent of concentration. There was no evidence of accumulation of ceftazidime in the serum in individuals with normal renal function following multiple IV doses of 1 and 2g every 8 hours for 10 days. Mean peak plasma concentrations of 17µg & 39µg per ml have been reported approximately one hour after intramuscular administration of 0.5 and 1g of ceftazidime, respectively. Five minutes after intravenous bolus injections of 0.5, 1, 2g of ceftazidime mean plasma concentrations of 45, 90, and 170µg per ml, respectively, have been reported. The plasma half-life is about 2 hours, but this is prolonged in patients with renal failure and in neonates. Clearance may be enhanced in patients with cystic fibrosis. Approximately 80 to 90% of an IM or IV dose of ceftazidime is excreted unchanged by the kidneys over a 24-hour period. After the IV administration of single 500mg or 1g doses, approximately 50% of the doses appeared in the urine in the first 2 hours. Additional 20% was excreted between 2 and 4 hours after dosing, and approximately another 12% of the dose appeared in the urine between 4 and 8 hours later. The elimination of ceftazidime by the kidneys resulted in high therapeutic concentrations in the urine. Ceftazidime is widely distributed in body tissue and fluids; therapeutic concentrations are achieved in the cerebrospinal fluid when the meninges are inflamed. It crosses the placenta and excreted in breast milk. Ceftazidime is passively excreted in bile, although only a small proportion is eliminated by this route. It is mainly excreted by the kidneys, almost exclusively by glomerular filtration; probenecid has little effect on the excretion. About 80 to 90% of a dose is recovered unchanged in the urine within 24 hours. It is removed by haemodialysis and peritoneal dialysis.

INDICATIONS:

Ceftazidime is a bactericidal cephalosporin antibiotic which is resistant to most beta-lactamases and is active against a wide range of Gram-positive and Gram-negative bacteria and indicated for the treatment of single infections and for mixed infections caused by two or more susceptible organisms. It may be used for injections caused by organisms resistant to other antibiotics including aminoglycosides and many cephalosporins.

Severe infections:

For example, septicaemia, bacteraemia, peritonitis, meningitis, infection in immunosuppressed patients with haematological or solid malignancies, and in patients in intensive care units with specific problems eg. infected burns.

Respiratory tract infections:

For example pneumonia, bronchopneumonia, infected pleurisy, empyema, lung abscess, infected bronchiectasis and bronchitis and in lung infections in patients with cystic fibrosis.

Ear, nose and throat infections:

For example, otitis media, malignant otitis externa, mastoiditis, sinusitis and other severe ear and throat infections.

Urinary tract infections:

For example, acute and chronic pyelonephritis, pyelitis, prostatitis, cystitis, urethritis (bacterial only), renal abscess and infections associated with bladder and renal stones.

Skin and soft tissues infections:

For example, erysipelas, abscesses, cellulitis, infected burns and wounds, mastitis, skin ulcers.

Gastrointestinal biliary and abdominal infections:

For example, cholangitis, cholecystitis, empyema of gall bladder, intra-abdominal abscesses, peritonitis, diverticulitis, post-partum and pelvic inflammatory conditions.

Bone and joints infections:

For example, osteitis, osteomyelitis, septic arthritis, infected bursitis.

Dialysis:

Infections associated with haemo- and peritoneal dialysis and with continuous ambulatory peritoneal dialysis (CAPD).

SIDE EFFECTS :

Ceftazidime is generally well tolerated. Adverse reactions are infrequent and include :

Local:

Thrombophlebitis with IV administration, pain and / or inflammation after IM injection.

Hypersensitivity:

Maculopapular or urticarial rash, fever, pruritus, and very rarely angioedema and anaphylaxis. (bronchospasm and / or hypotension).

Gastrointestinal:

Diarrhoea, nausea, vomiting, abdominal pain and very rarely oral thrush or colitis. As with other cephalosporins, colitis may be associated with *Clostridium difficile* and may be present as pseudomembranous colitis.

Genito-urinary:

Candidiasis, vaginitis.

Central nervous system:

Headache, dizziness, paraesthesia and bad taste. A few reports of neurological sequelae including tremor, myoclonia, convulsions, and encephalopathy in patients with renal impairment in whom the dose of Ceftazidime has not been appropriately reduced. Laboratory test changes noted transiently during Ceftazidime therapy include: the hepatic enzymes, ALT (SCOT), AST (SCPT), LDH, GGT, and alkaline phosphatase. As with some other cephalosporins, transient elevation of blood urea, blood urea nitrogen and / or serum creatinine have been observed occasionally. Very rarely, leukopenia, neutropenia, agranulocytosis, thrombocytopenia and lymphocytosis have been reported.

WARNINGS AND PRECAUTIONS :

Cephalosporin antibiotics at high dosage should be given with caution to patients receiving concurrent treatment with nephrotoxic drugs, e.g. aminoglycoside antibiotics, or potent diuretics such as frusemide, as these combinations are suspected of affecting renal function adversely. Clinical experience with Ceftazidime has shown that this is not likely to be a problem at the recommended dose levels. There is no evidence that ceftazidime adversely affects renal function at normal therapeutic doses: Ceftazidime is eliminated via kidneys, therefore the dosage should be reduced according to the degree of renal impairment. When the dose has not been reduced appropriately, neurological sequelae has been reported occasionally. As with other broads spectrum antibiotics, prolonged use of Ceftazidime may result in overgrowth of non-susceptible organisms (eg. Candida, Enterococci) which may require interruption of treatment or adoption of appropriate measures.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. As with other beta-lactam antibiotics, before initiating therapy with Vaxcel Ceftazidime Injection, careful inquiry should be made concerning previous hypersensitivity reactions to Ceftazidime, penicillins, cephalosporins, carbapenems or other beta-lactam agents. Ceftazidime should be given only with special caution to patients with type 1 or immediate hypersensitivity reaction to penicillin. If an allergic reaction occurs, Vaxcel Ceftazidime Injection must be discontinued immediately and appropriate alternative therapy instituted. Serious hypersensitivity reactions may require epinephrine, hydrocortisone, antihistamine or other emergency measures.

Pregnancy and Lactations:

There is no experimental evidence of embryopathic or teratogenic effects attributable to ceftazidime but, as with all drugs, it should be administered with caution during the early months of pregnancy and in early infancy. Use in pregnancy requires that the anticipated benefit be weighed against the possible risks. Since ceftazidime is excreted in human milk with low concentration, consequently caution should be exercised when ceftazidime is administered to a nursing mother.

CONTRAINDICATIONS :

Patients known to be hypersensitive to Cephalosporin antibiotics.

DRUG INTERACTIONS :

As Cephalosporin antibiotics, Ceftazidime at high dosage should be given with caution to patients receiving concurrent treatment with nephrotoxic drugs, eg. aminoglycoside antibiotics, or potent diuretics such as frusemide, as these combinations are suspected of affecting renal function adversely. Clinical experience with ceftazidime has shown that this is not likely to be problem at the recommended dose levels. Chloramphenicol is antagonistic in vitro with ceftazidime and other cephalosporins. The clinical relevance of this finding is unknown, but if concurrent administration of ceftazidime with chloramphenicol proposed, the possibility of antagonism should be considered.

DOSAGE :

General dosage recommendations

Ceftazidime is to be used by the parenteral route, the dosage depending upon the severity, sensitivity and type of infection and the age, weight and renal function of the patient.

Adults

The adults dosage range for Ceftazidime is 1 to 6g per day : for instance, 500mg, 1g or 2g given 12 or 8 hourly by IV or IM injection. In urinary tract infections and in many less serious infections, 500mg or 1g 12 hourly is usually adequate. In the majority of infections, 1g 8 hourly or 2g 12 hourly should be given. In very severe infections, especially in immunocompromised patients, including those with neutropenia, 2g, 8 or 12 hourly should be administered.

Cystic fibrosis

In fibrocystic adults with normal renal function who have pseudomonas lung infections, high doses of 100 to 150mg/kg/day as three divided doses should be used. In adults with normal renal function 9g/day has been used safely.

Infants and children

The usual dosage range for children aged over two months is 30 to 100mg/kg/day, given as two or three divided dosage. Doses up to 150mg/kg/day (maximum 6g daily) in three divided doses may be given to infected immunocompromised or fibrocystic children or children with meningitis.

Neonates and children up to 2 months of age

A dose of 25 to 60 mg/kg/day given as two divided doses has been proved to be effective. In the neonate the serum half-life of Ceftazidime can be three to four times that in adults.

Uses in elderly

In view of the reduced clearance of ceftazidime in acutely ill elderly patients, the daily dosage should not normally exceed 3g, especially in those over 80 years of age.

Dosage in Impaired Renal Function

Ceftazidime is excreted by the kidneys almost exclusively by glomerular filtration. Therefore, in patients with impaired renal function it is recommended that the dosage of Ceftazidime should be reduced to compensate for its slower excretion, except in mild impairment. In patients with suspected renal insufficiency, an initial loading dose of

1gm of Ceftazidime may be given. An estimate of GFR should be made to determine the appropriate maintenance dose.

Recommended maintenance doses are shown below:

RECOMMENDED MAINTENANCE DOSES OF CEFTAZIDIME IN RENAL INSUFFICIENCY

Creatinine Clearance ml/min	Approx. serum creatinine* µmol/L (mg/dl)	Recommended unit dose of Ceftazidime (g)	Frequency of dosing (hourly)
> 50	< 150	Normal dosage	Normal dosage
50 - 31	150 - 200 (1.7 - 2.3)	1.0	12
30 - 16	200 - 350 (2.3 - 4.0)	1.0	24
15 - 6	350 - 500 (4.0 - 5.6)	0.5	24
≤ 5	> 500 (> 5.6)	0.5	48

* these values are guidelines and may not accurately predict renal function in all patients especially in the elderly in whom the serum creatinine concentration may overestimate renal function.

In patients with severe infections, especially in neutropenics, who would normally receive 6g of ceftazidime daily were it not for renal insufficiency, the unit dose given may be increased by 50% or the dosing frequency increased appropriately. In such patients it is recommended that ceftazidime serum levels should be monitored and trough levels should not exceed 40mg/litre. When only serum creatinine is available, the Cockcroft's equation may be used to estimate creatinine should represent a steady state of renal function:

Males

$$\text{Creatinine clearance (ml/min)} = \frac{\text{Weight (kg)} \times (140) - (\text{age in years})}{72 \times \text{serum creatinine (mg/dl)}}$$

Females

0.85 x above value. (To convert serum creatinine in µmol/litre into mg/dl divide by 88.4). In children the creatinine clearance should be adjusted for body surface area or lean body mass and the dosing frequency reduced in cases of renal insufficiency as for adults.

The serum half-life of ceftazidime during haemodialysis ranges from 3 to 5 hours. The appropriate maintenance dose of ceftazidime should be repeated following each haemodialysis period.

Dosage in peritoneal dialysis

Ceftazidime may also be used in peritoneal dialysis and continues ambulatory peritoneal dialysis (CAPD). As well as using ceftazidime intravenously, it can be incorporated

into the dialysis fluid (usually 125 to 250 mg for 2L of dialysis fluid).

Administration

Ceftazidime may be given intravenously or by deep intramuscular injection into a large muscle mass such as the upper outer quadrant or the gluteus maximus or lateral part of the thigh.

SYMPTOMS AND TREATMENTS FOR OVERDOSAGE :

Overdosage can lead to neurological sequelae including encephalopathy, convulsions and coma. Serum levels of Ceftazidime can be reduced by haemodialysis or peritoneal dialysis.

CAUTIONS FOR USAGE:

This product do not contain any preservatives and should be used as a single-dose preparations. In keeping with good pharmaceutical practice is preferable to use freshly constituted solutions of Ceftazidime for injections. If this is not practicable, satisfactory potency is retained for 24 hours at room temperature (below 25°C) when prepared in Water for injections BP or any of the injections listed as follows.

STORAGE :

Keep the container well closed. Store below 30°C. Protect from light. Freshly reconstituted solution is recommended.

DIRECTION FOR RECONSTITUTION:

Diluent for reconstitution:

- At Ceftazidime concentrations between 1mg/ml and 40mg/ml in : 0.9% Sodium Chloride Injection BP, Compound Sodium Lactate Injection BP (Hartmann's Solution), 5% Dextrose Injection BP, 0.225% Sodium Chloride and 5% Dextrose Injection BP, 0.45% Sodium Chloride and 5% Dextrose Injection BP, 0.9% Sodium Chloride and 5% Dextrose Injection BP, 0.18% Sodium Chloride and 4% Dextrose Injection BP, 10% Dextrose Injection BP.

- When reconstituted for IM use with : 0.5% or 1% Lignocaine HCl Injection BP.

- When admixed at 4mg/ml with: (both component retain satisfactory potency), Hydrocortisone (hydrocortisone sodium phosphate) 1mg/ml in 0.9% Sodium Chloride Injection BP or 5% Dextrose Injection BP, Cefuroxime (cefuroxime sodium) 3mg/ml in 0.9% Sodium Chloride Injection BP, Cloxacillin (cloxacillin sodium) 4mg/ml in 0.9% Sodium Chloride Injection BP, Heparin 10µg/ml or 50µg/ml in 0.9% Sodium Chloride Injection BP, Potassium Chloride 10mEq/L or 40mEq/L in 0.9% Sodium Chloride Injection BP.

- The contents of a 500mg vial for Ceftazidime, reconstituted with 1.5ml of Water for Injection, may be added to metronidazole injection (500mg in 100ml) and both retain their activity.

Incompatibilities:

Ceftazidime is less stable in sodium bicarbonate injection than other IV fluids. It is not recommended as diluent. Ceftazidime and aminoglycosides should not be mixed in the same giving set or syringe.

INSTRUCTION FOR RECONSTITUTION:

Preparation for Solution

Vial size		Amount of Diluent to be added (ml)	Approximate Concentration (mg/ml)
1g	IM	3.0	260
	IV bolus	10.0	90
	IV infusion	50.0*	20
2g	IV bolus	10.0	170
	IV infusion	50.0*	40

*Note: Addition should be in two stages (as below).

As the product dissolves, carbon dioxide is released and a positive pressure develops. For ease of use, it is recommended that the following techniques of reconstitution are adopted.

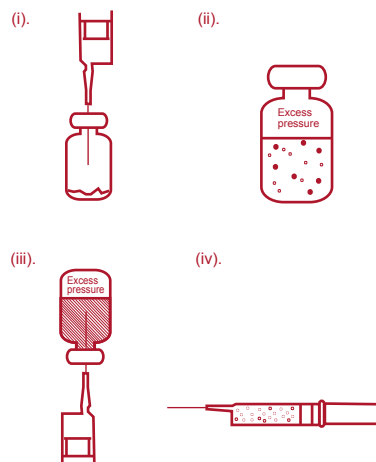
a) 1g (IM/IV) & 2g (IV bolus).

(i) Insert the syringe needle through the vial closure and inject the recommended volume of diluent. Remove the syringe needle.

(ii) Shake to dissolve: carbon dioxide is released and a clear solution obtained in about 1 to 2 minutes.

(iii) Invert the vial. With the syringe plunger fully depressed, insert the needle through the vial closure and withdraw the total volume of solution into the syringe (the pressure in the vial may aid withdrawal). Ensure that the needle remains within the solution and does not enter the head space.

(iv) The withdrawn solution may contain small bubbles of carbon dioxide; they may be disregarded.



b) 1g & 2g iv infusion. (eg. up to 30 minutes)

(i) Insert the syringe needle through the vial closure and inject 10ml of diluent. Then, remove the syringe needle.

(ii) Shake to dissolve: carbon dioxide is released and a clear solution obtained in about 1 to 2 minutes.

(iii) Insert a gas relief needle through the vial closure to relieve the internal pressure and with the gas relief in place, add the remainder of diluent. Remove the gas and syringe needle; shake the vial and set up for infusion use in the normal way.

Note: to preserve product sterility, it is important that a gas relief needle is NOT inserted through the vial closure before the product has dissolved. These solutions may be given directly into the vein or introduced into the tubing of a giving set if the patient is receiving parenteral fluids. Ceftazidime is compatible with the most commonly used intravenous fluids. Solutions range from light yellow to amber depending on concentration, diluent and storage conditions used. Within the stated recommendations, product potency is not adversely affected by such color variations.

PACK QUANTITIES:

Single vial per box and 10 vials per box.

Vaxcel Ceftazidime 1g Injection : MAL20020152AZ

Vaxcel Ceftazidime 2g Injection : MAL20020153AZ

Further information can be obtained from pharmacist, physician or the manufacturer.



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